

Federated Learning Approach for Obstructive Sleep Apnea Prediction

Bachelor thesis presentation

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Outline

(5, 3') Introduction

(3, 1') Project Background

(4, 2') Objective within the Bachelor Project

(6, 3') Research and/or Development

(2, 1') Solution

(3, 1') Project Approach

(2, 1') Output Demo

(4, 2') Reflection

Introduction

- ▶ Team
 - ▶ Manja Gorenc Novak, *Head of AI and Senior Data Scientist*
 - ▶ Rok Vinder, *Data Analyst and Machine Learning Engineer*
- ▶ Result d.o.o
 - ▶ Slovenian Company
 - ▶ ~ 50 employees
 - ▶ Software development
 - ▶ Data and AI
 - ▶ Business solutions
- ▶ Context
 - ▶ eEarly
 - ▶ Health data

Project background

Problem statement

Stakeholders and needs

- ▶ **Developers**

- ▶ Understand code
- ▶ Interpret results
- ▶ Run the project

- ▶ Patients

- ▶ Data privacy

- ▶ Doctors

- ▶ Improved diagnosis
- ▶ Better tools

- ▶ (Lawyers)

- ▶ Less regulation burden

Objective within the Bachelor Thesis

- ▶ Proposed solution
- ▶ MVP
- ▶ Broader scope

Research and/or development

- ▶ Comparative study
- ▶ Technology decision

Solution

Solution

- Final solution

Project Approach

Project Approach

- ▶ Development method
- ▶ MVP and extras
- ▶ Collaboration

Output and Demo

Reflection

Reflection

- ▶ Choice reflection
- ▶ Achieved objectives
- ▶ Improvements and weaknesses
- ▶ Added value
- ▶ Limitations
- ▶ Impact
 - ▶ People
 - ▶ Environment
 - ▶ Research
 - ▶ Healthcare